Generative Machine Learning Models for Data Assimilation

Polymath Jr.

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Abstract

We study normalizing flows from the perspective of optimal transport, drawing connections between Monge's formulation, Kantorovich's formulation, and Brenier's theorem: implementing these ideas in a practical mesh-free setting. In the quadratic cost case, Brenier's theorem ensures that the optimal transport map can be expressed as the gradient of a convex potential. Following the update rule established in the existing literature, we consider two parameterizations of this potential. The first is a radial basis function approach that departs from prior work in two key aspects: the gradient of the potential is evaluated at the original source sample point rather than at the current transported point, and convexity is enforced throughout the flow using a linear program derived from Farkas' Lemma. The second is an Input Convex Neural Network (ICNN) parameterization, which guarantees convexity by construction and therefore requires no LP-based enforcement. Experiments on synthetic two-dimensional datasets demonstrate that maintaining convexity in the potential improves stability, interpretability, and theoretical consistency of the learned maps.

Monge and Kantorovich Formulations

In optimal transport, Monge's original problem seeks a map $y : \mathbb{R}^n \to \mathbb{R}^n$ that transports a source density $\rho(x)$ to a target density $\mu(y)$ while minimizing a cost functional

$$M(y) = \int_{\mathbb{R}^n} c(x, y(x))\, \rho(x)\, dx.$$

This formulation enforces a one-to-one mapping between source and target points and is constrained by the change-of-variables relation $\rho(x) = J^y(x)\, \mu(y(x))$, where $J^y(x) = \det(\nabla y(x))$ is the Jacobian determinant. Kantorovich's relaxation replaces this rigid map with a joint distribution $\pi(x, y)$ over $\mathbb{R}^n \times \mathbb{R}^n$ that satisfies the marginal constraints

$$\int \pi(x, y)\, dy = \rho(x), \qquad \int \pi(x, y)\, dx = \mu(y).$$

The problem becomes minimizing

$$\min_{\pi} \int_{\mathbb{R}^n \times \mathbb{R}^n} c(x, y)\, \pi(x, y)\, dx\, dy,$$

and admits a dual formulation:

$$\max_{u,v} \left[\int u(x)\rho(x)\, dx + \int v(y)\mu(y)\, dy \right]$$

subject to $u(x) + v(y) \leq c(x, y)$ for all x, y.

Brenier's Theorem

Let ρ and μ be two absolutely continuous probability densities on $\mathbb{R}^d$ with finite second moments. Then there exists a **unique** optimal transport map

$$T^* : \mathbb{R}^d \to \mathbb{R}^d$$

pushing ρ forward to μ. Moreover, this map is the gradient of a convex potential function ϕ:

$$T^*(z) = \nabla \phi(z),$$

where $\phi : \mathbb{R}^d \to \mathbb{R}$ is convex, and $\nabla \phi(z)$ specifies the destination of the mass originally at z. Under **quadratic cost**, the optimal way to transport mass is to follow the gradient of this convex potential.

Mesh-Free Flow Implementation

We work in a mesh-free setting where the source and target densities are known only via samples $x_i \sim \rho$ and $y_j \sim \mu$ for $i = 1, \dots, m$ and $j = 1, \dots, n$. In the case of quadratic cost $c(x, y) = \frac{1}{2}\|x - y\|^2$, Monge and Kantorovich are connected via

$$T(x) = \nabla U(x),$$

where U is a convex potential (often called the Brenier potential). The Kantorovich dual potentials can then be expressed as

$$u(x) = \tfrac{1}{2}\|x\|^2 - U(x), \qquad v(y) = \tfrac{1}{2}\|y\|^2 - U^*(y).$$

Let $z^k(x)$ denote the iterative map at step k, initialized by $z^0(x) = x$ and updated via

$$z_i^{k+1} = \nabla U_\beta^k(z_i^k),$$

where

$$U_\beta^k(z) = \tfrac{1}{2}\|z\|^2 + \sum_\ell \beta_\ell F_\ell^k(z), \quad \nabla U_\beta^k(z) = z + \sum_\ell \beta_\ell \nabla F_\ell^k(z).$$

As the flow evolves, the map converges:

$$\lim_{k \to \infty} z^k(x) = y(x).$$

The coefficients β_ℓ are fitted by minimizing a Monte Carlo dual functional:

$$\hat{D}^k(\beta) = \frac{1}{m}\sum_{i=1}^{m} U_\beta^k(z_i^k) + \frac{1}{n}\sum_{j=1}^{n} \left(U_\beta^k\right)^*(y_j),$$

through gradient updates

$$\bar{\beta}_\ell \leftarrow \bar{\beta}_\ell - \eta \left[\frac{1}{m}\sum_{i=1}^{m} F_\ell^k(z_i^k) - \frac{1}{n}\sum_{j=1}^{n} F_\ell^k(y_j) \right].$$

We choose radially symmetric basis functions around centers $\tilde{z}_\ell$:

$$F_\ell(z) = r\, \mathrm{erf}\!\left(\frac{r}{\alpha}\right) + \frac{\alpha}{\sqrt{\pi}} e^{-(r/\alpha)^2}, \quad r = \|z - \tilde{z}_\ell\|,$$

Bandwidth α depends on local densities:

$$\alpha \propto \left(\frac{n_p}{n+m} \left(\frac{1}{\bar{\rho}(\tilde{z}_l)} + \frac{1}{\bar{\mu}(\tilde{z}_l)} \right) \right)^{1/d}.$$

This yields explicit particle updates:

$$z^{k+1} = z^k + \sum_\ell \beta_\ell f_\ell(r)\, (z^k - \tilde{z}_\ell), \quad f_\ell(r) = \frac{\mathrm{erf}(r/\alpha)}{r}$$

where the centers $\tilde{z}_\ell$ are re-chosen at each step to localize updates. After k steps:

$$z^k(x) = x + \mathrm{pert}_0 + \mathrm{pert}_1 + \cdots + \mathrm{pert}_{k-1}$$

Enforcing Convexity with Linear Programming

We study when convexity is preserved in the update

$$U_\beta^{k+1}(z) = U_\beta^k(z) + \beta_{k+1} F_{k+1}(z),$$

where U_β^k and F_{k+1} are convex. If all $\beta_l \geq 0$, convexity is guaranteed. When some $\beta_l < 0$, convexity must be checked explicitly. From the interpolation condition

$$U_\beta^{k+1}(z_0^{(j)}) \;\geq\; U_\beta^{k+1}(z_0^{(i)}) + g_i^\top (z_0^{(j)} - z_0^{(i)}) \quad \forall j,$$

convexity holds if these inequalities are feasible for g_i. By Farkas' Lemma, infeasibility of these inequalities for some i is equivalent to the existence of $\alpha \geq 0$ such that

$$\sum_{j=1}^{m} \alpha_j (z_0^{(j)} - z_0^{(i)}) = 0, \quad \sum_{j=1}^{m} \alpha_j \big(U_\beta^{k+1}(z_0^{(j)}) - U_\beta^{k+1}(z_0^{(i)})\big) < 0.$$

Since LP solvers cannot handle strict inequalities, we instead solve

$$\min_{\alpha \geq 0} \sum_{j=1}^{m} \alpha_j \big(U_\beta^{k+1}(z_0^{(j)}) - U_\beta^{k+1}(z_0^{(i)})\big) \quad \text{s.t.} \quad \sum_{j=1}^{m} \alpha_j (z_0^{(j)} - z_0^{(i)}) = 0.$$

If the LP is unbounded for any i, the inequalities are infeasible and U_β^{k+1} is not convex. In practical terms, if this condition holds, there is no affine function that acts as a valid lower local linear approximation to U_β^{k+1} at $z_0^{(i)}$. Geometrically, the supporting hyperplane condition fails at that point. Since any convex function can be represented as the pointwise maximum of its supporting affine functions, the absence of such a function means that no convex function can interpolate through these values at $z_0^{(i)}$.

Divergence from Published Literature

Our method departs from the published literature in two functional ways. The first, we use ICNN components to enforce convexity, parameterizing the potential as

$$T(x) = \nabla \phi_\alpha(x) = \alpha x + \sum_{k=1}^{K} \beta_k \nabla F_k(x),$$

with each F_k implemented as an *Input Convex Neural Network (ICNN)*. The second, we use a special localized elementary map while enforcing convexity with LP:

$$X_i^{n+1} = X_i^n + B\, \nabla_{X^0} F\left(r_i^{n+1}\right) = x_i^n + \beta\, f\left(r_i^{n+1}\right) \left(x_i^0 - \hat{x}^{n+1}\right)$$

where

$$r_i^{n+1} = \|X_i^0 - \hat{X}^{n+1}\|, \qquad f(r) = \frac{1}{r}\frac{dF(r)}{dr}.$$

References

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